CODI-GPT-2 · watching computation across latent depth

Goal: understand HOW operator and number representations interact during the 6 latent thought steps and the final decode.

Setup:

- Activations: 1000 SVAMP problems × (latent_steps=6, layers=13, hidden=768)
- Probes fit on the residual stream at every (layer, latent_step).

Three views:

- (1) variable probes — what the residual ENCODES at each layer
- (2) logit lens (CODI lm_head + ln_f) — what the model would PREDICT
  caveat: at decode-position-1 the model emits 'The...', not the answer.
- (3) candidate-rank trajectory across the 4 operator answers

(1a) Variable probes · last prompt token

Setup

Per-(layer, latent_step) logistic regression on (residual, label).

Three labels:

- units digit of gold answer (10-class, chance 10%)
- tens digit of gold answer (10-class, chance 10%)
- operator type (4-class, chance 25%)

Peak accuracies

- operator: 91.0% (chance 25%)
- tens digit: 45.5%
- units digit: 26.0%

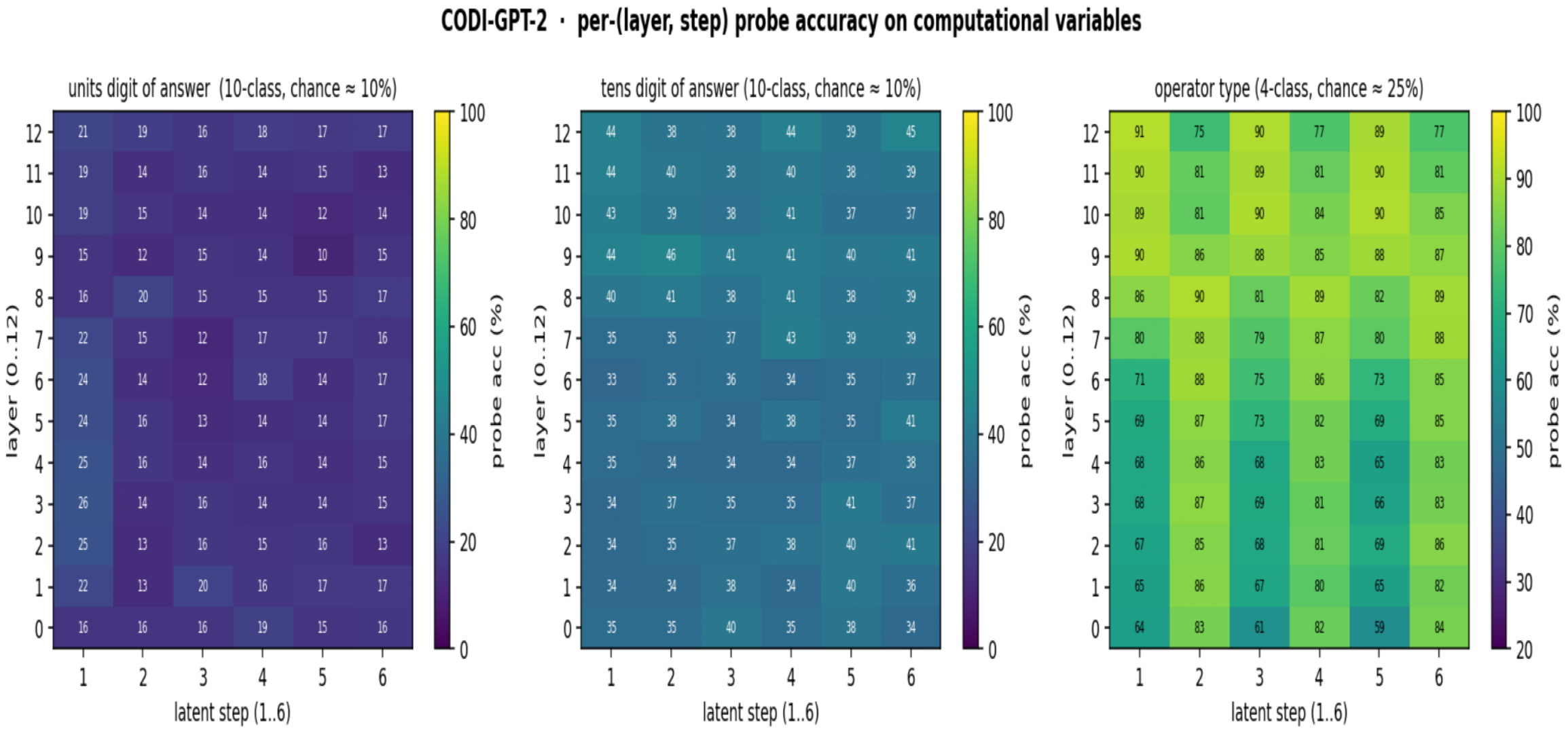
Reading

Operator is cleanly encoded across nearly all layers/steps.

Tens digit is encoded ~4× chance, units digit barely above chance.

Consistent with model getting the right OPERATOR but only roughly the right NUMBER.

(1a) Variable probes heatmap (last prompt token)



operator probe is sharp and broad; digit probes are weak. Probing the residual at the LAST prompt token, before EOT.

(1b) Variable probes · answer position (post-EOT, decode forward)

layer	units digit	tens digit	operator
L0	14.5%	40.0%	53.0%
L1	17.0%	39.0%	77.0%
L2	14.0%	36.5%	73.5%
L3	16.0%	36.5%	76.0%
L4	17.0%	36.5%	74.5%
L5	17.5%	37.5%	77.5%
L6	14.5%	37.0%	77.5%
L7	14.0%	36.5%	79.0%
L8	15.5%	35.0%	83.5%
L9	15.5%	32.5%	85.5%
L10	14.0%	36.5%	86.0%
L11	15.5%	37.5%	86.5%
L12	16.0%	41.5%	82.5%
PEAK	17.5% @L5	41.5% @L12	86.5% @L11

Logistic regression on the residual at the FIRST decode forward (immediately after EOT). Same pattern as last-prompt-token: operator strong, digit probes weak.

(2) Logit lens · caveat · WHY top-1 = 0% at decode-position-1

What we did

- Loaded CODI-GPT-2's actual lm_head (50260 × 768) and final layer norm (ln_f).
- For each example at every layer of the first decode forward, computed argmax(ln_f(residual) @ W^T) and compared to the gold answer's first token.

Result

- 0% top-1 = gold answer at every layer.

Why

- Inspecting the actual top-1 token at L12: '"The"' on 999/1000 examples.
- The model's output template is 'The answer is: <number>'. So the FIRST decode forward predicts 'The' with very high probability — NOT the number.
- The numeric answer is only emitted several tokens later, mid-generation.

Implication

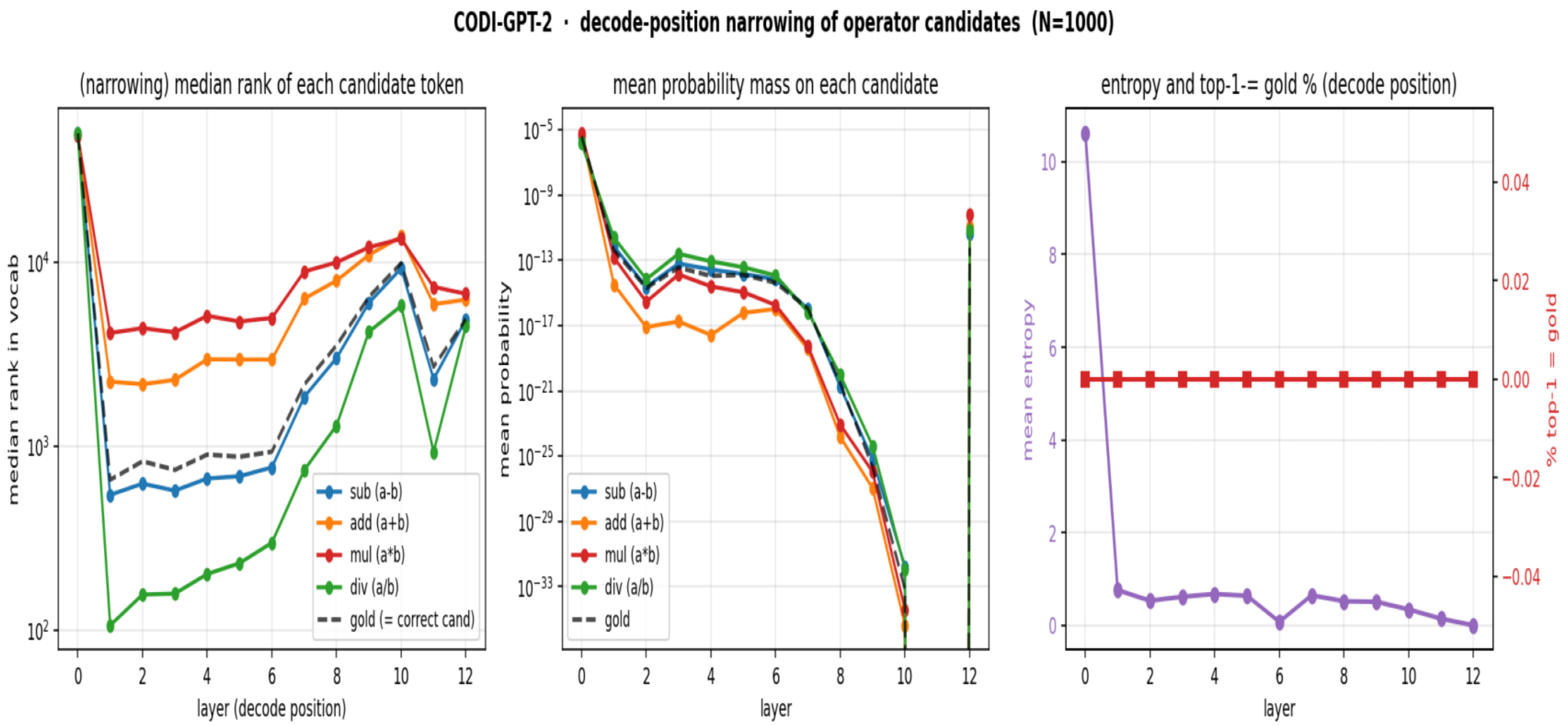
- Logit-lens / candidate-rank measured at decode-position-1 is reading the TEMPLATE position, not the answer position. The 'narrowing-down' interpretation doesn't apply there.
- A clean version requires capturing activations at the position where the model is actually writing the answer NUMBER (mid-generation). Not done in this run.

What still IS valid here

- The variable probes (slides 1a/1b) — they probe what the RESIDUAL ENCODES, independent of which token comes out next.
- The relative cos_sim trajectories from the broader number-isolation analysis.

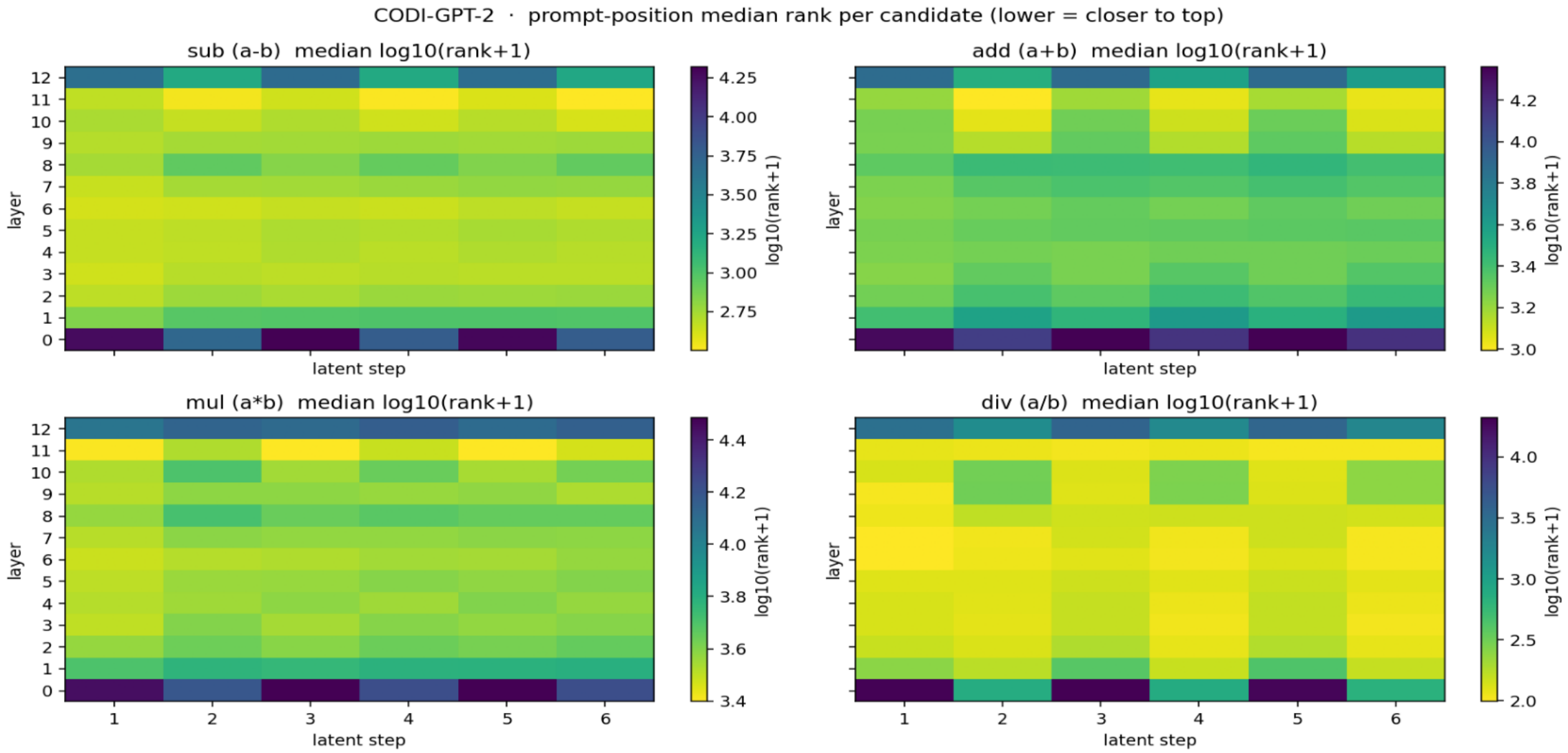
known issue, see svamp_decode_preds.json — first decoded token is overwhelmingly 'The'

(3) Candidate-rank trajectory · decode position-1 (caveat: template token)



ens at decode position-1. DIV consistently lowest rank because small-valued tokens are inherently more probable in the model's prior, NOT because the model is choosing division. Reading is dominated by the ten

(3b) Candidate-rank trajectory · last prompt token (per layer × step)



e caveat as (3): residual at last prompt token isn't the model's answer commitment, it's the input to the latent loop. Useful for comparing relative trajectories of the 4 operator candidates, NOT for absolute predict

Summary

What we can confidently say

- Operator info is cleanly encoded throughout latent depth. Probe accuracy is 91.0% (last prompt) and 86.5% (decode position).
- Specific digit values are only weakly encoded. Tens digit ~40-45% (chance 10%), units digit ~17-26%. The model commits firmly to OPERATOR but only roughly to VALUE.

What we cannot say

- We cannot 'watch addition happen' from these probes — the logit-lens sees the template token at decode-position-1, not the answer position.

What would answer the question

- Capture activations at the position where the actual answer NUMBER is being emitted (mid-decode, ~position 4-6 of 'The answer is: <num>').
- Re-run logit lens at that position; track candidate ranks per layer there.
- Per-position digit-classifier trained at each token slot might also work.